

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS – GRADE 10 | SUB-STRAND 2.9: LINEAR MOTION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This Final Explanation is your opportunity to show everything you have learned about linear motion by answering the driving question fully and completely. You will use mathematics — both calculations and graphs — to tell the complete story of motion, just like a sports analyst breaking down a Nairobi Marathon race.

READ THE DRIVING QUESTION CAREFULLY BEFORE YOU BEGIN:

"How can we use mathematics — calculations and graphs — to fully describe and explain the motion of any moving body, from a Nairobi Marathon runner to a matatu on the Nakuru highway?"

YOUR TASK: Write a detailed, structured explanation that answers the driving question. Your response must include all four sections below. For each section, write in full sentences, show all working for any calculations, and draw or describe graphs clearly with labelled axes, units, and key features. Use specific values and examples — do not give vague or general answers.

BEFORE YOU WRITE, ask yourself:

- Can I explain the difference between distance and displacement, speed and velocity?
- Can I calculate average speed, average velocity, and read instantaneous values from a graph?
- Can I sketch AND interpret both a displacement-time graph and a velocity-time graph?
- Can I explain what the shape of each graph tells us — not just the start and end points?
- Can I connect every idea back to a real moving body (the marathon runners, a matatu, an athlete)?

You have 60 minutes. Write directly in the spaces provided under each section.

Section 1 — The Language of Motion: Key Concepts and Definitions

Define and distinguish between the following pairs of terms, using the Nairobi Marathon

EXEMPLAR RESPONSE:

runners OR a matatu on the Nakuru highway as your context. For each definition, include the formula, SI units, and ONE worked numerical example using realistic values. (a) Distance vs. Displacement (b) Speed vs. Velocity (c) Average speed/velocity vs. Instantaneous speed/velocity (d) Uniform motion vs. Non-uniform motion (acceleration) Finish this section by explaining: Why is it NOT enough to say 'Runner B covered 42 km in 2 hours 12 minutes' if you want to fully understand how Runner B moved during the race?	(a) Distance vs. Displacement Distance is the total length of the path travelled by a moving body, regardless of direction. It is a scalar quantity — it has magnitude only. Displacement is the straight-line distance from the starting point to the finishing point, measured in a specific direction. It is a vector quantity — it has both magnitude and direction. Marathon context: The 2023 Nairobi Marathon course winds through Uhuru Highway, Haile Selassie Avenue, and back. Runner A travels a total distance of 42.195 km along the road. However, if the start and finish line are at the same point (as in a loop course), the displacement is 0 km. If the course is a straight out-and-back on Mombasa Road, and the finish is 10 km north of the start, displacement = 10 km north. (b) Speed vs. Velocity Speed = Distance ÷ Time. It is a scalar quantity (magnitude only). SI unit: metres per second (m/s). Velocity = Displacement ÷ Time. It is a vector quantity (magnitude + direction). SI unit: m/s. Worked example — Nakuru highway matatu: A matatu travels 90 km from Nairobi to Nakuru in 1.5 hours, heading north-west. Average speed = $90 \text{ km} \div 1.5 \text{ h} = 60 \text{ km/h} = 16.7 \text{ m/s}$ Average velocity = $90 \text{ km north-west} \div 1.5 \text{ h} = 60 \text{ km/h north-west} = 16.7 \text{ m/s north-west}$ (c) Average speed/velocity vs. Instantaneous speed/velocity Average speed is the total distance divided by the total time for an entire journey. Instantaneous speed is the speed of a body at one specific moment in time — the value you would read on a speedometer at a single instant. Worked example — Runner B: Runner B's split time at the 35 km mark is 1 h 58 min, and at the 40 km mark it is 2 h 06 min. Average speed over that 5 km segment = $5 \text{ km} \div 8 \text{ min} = 0.625 \text{ km/min} = 10.4 \text{ m/s} \dots$ [Re-calculation]: $5000 \text{ m} \div 480 \text{ s} = 10.4 \text{ m/s}$. This tells us Runner B's average speed in that segment, but the instantaneous velocity at exactly the 37 km mark could be higher or lower — we cannot know from the table alone. That is why we need a graph. (d) Uniform vs. Non-uniform motion Uniform motion: a body moves equal distances in equal time intervals — speed is constant, acceleration = 0. Runner A's middle kilometres (km 10–35) are approximately uniform. Non-uniform motion: a body covers unequal distances in equal time intervals — speed is changing, meaning the body is accelerating or decelerating. Runner B shows non-uniform motion: fast start, slower middle, sprint finish. Why a single total-time statement is not enough: Saying 'Runner B covered 42 km in 2 h 12 min' gives us only one number — the overall average speed $\approx 19.2 \text{ km/h}$. It tells us nothing about the fact that Runner B was running at perhaps 22 km/h in the first 5 km, dropped to 17 km/h by km 25, then surged to 23 km/h in the final 2 km. The single number hides the entire story of acceleration, strategy, and the critical sprint that Runner A used to overtake Runner B. Only split-time data and graphs reveal that full story.
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Section 2 — Displacement-Time Graphs: Drawing, Reading, and Interpreting

Using the split-time data below for Runner A and Runner B, answer all parts.

SPLIT-TIME TABLE
(approximate values for use in this assessment):

| Distance (km) | Runner A time (min) | Runner B time (min) |

0	0	0
5	22	19
10	44	40
15	66	63
20	88	87
25	110	113
30	132	138
35	154	163
40	176	186
42	185	196

(a) Sketch a displacement-time graph for BOTH runners on the same axes. Label axes with correct quantities and units. Plot all data points and draw a smooth curve through them.

(b) What does the GRADIENT (slope) of a displacement-time graph represent? State the formula and units.

EXEMPLAR RESPONSE:

(a) Graph Description (sketch in your booklet):

Horizontal axis (x-axis): Time in minutes, labelled 'Time (min)', scale 0 to 200 min.

Vertical axis (y-axis): Displacement in km, labelled 'Displacement (km)', scale 0 to 42 km.

Runner A: Points are nearly equally spaced in time — the curve is close to a straight line (nearly uniform gradient), indicating nearly constant speed throughout.

Runner B: The first segment (0–10 km) has a steeper gradient than Runner A — Runner B's curve rises more quickly, showing higher speed. From km 15–35, the gradient becomes shallower than Runner A — Runner B slows. From km 40–42, the gradient steepens again — the sprint finish. The two curves cross between km 30 and km 35.

(b) Gradient of a displacement-time graph = Velocity

Formula: $\text{gradient} = \Delta \text{displacement} \div \Delta \text{time} = (s_2 - s_1) \div (t_2 - t_1)$

Units: km/min or m/s

A steeper gradient → higher velocity. A shallower gradient → lower velocity. A horizontal line → the body is stationary (velocity = 0). A straight line → uniform velocity. A curve → changing velocity (acceleration or deceleration).

(c) Average speed calculations:

Runner A: Total distance = 42 km, Total time = 185 min = $185 \div 60 \text{ h} = 3.083 \text{ h}$

Average speed = $42 \div 3.083 = 13.62 \text{ km/h} = 13620 \text{ m} \div 11100 \text{ s} = 3.78 \text{ m/s} \checkmark$

Runner B: Total distance = 42 km, Total time = 196 min = 3.267 h

Average speed = $42 \div 3.267 = 12.86 \text{ km/h} = 3.57 \text{ m/s} \checkmark$

Runner A's average speed is higher — confirming Runner A won, despite being behind at km 30.

(d) Three key differences identified from the graph:

① km 0–10 (time 0–40 min for A, 0–40 min for B):

Runner B's curve has a noticeably steeper gradient in the first 5 km (Runner B reaches km 5 at 19 min vs Runner A at 22 min).

Runner B was running faster — approximately $5 \text{ km} \div 19 \text{ min} = 0.263 \text{ km/min} = 15.8 \text{ km/h}$ compared to Runner A's $5 \text{ km} \div 22 \text{ min} = 13.6 \text{ km/h}$. Runner B's fast start is clear from the curve being above and to the left of Runner A's.

② km 20–35 (the middle section):

Here Runner B's gradient becomes shallower than Runner A's — Runner B's curve flattens relative to Runner A's. For example, km 25–30: Runner B takes $138 - 113 = 25 \text{ min}$ for 5 km → speed = $5 \div 25 = 0.20 \text{ km/min} = 12 \text{ km/h}$. Runner A takes $132 - 110 = 22 \text{ min}$ for 5 km → speed = $5 \div 22 = 0.227 \text{ km/min} = 13.6 \text{ km/h}$. Runner A is pulling ahead in speed, even though Runner B is still ahead in position at this point.

(c) Calculate the average speed of Runner A and Runner B over the full 42 km. Show all working. (d) Using the graph, identify and explain THREE specific moments or segments where the two runners' speeds were different. Refer to specific km marks and describe what the shape of each curve tells you. (e) Runner B was ahead at km 30. Use the graph to explain precisely how and when Runner A overtook Runner B. What does the crossing of the two curves represent?	③ km 40–42 (the sprint finish): Runner B's gradient steepens sharply again — Runner B sprints. Runner B covers the last 2 km in $196 - 186 = 10$ min $\rightarrow$ speed = $2 \div 10 = 0.2$ km/min = 12 km/h. But Runner A covers it in $185 - 176 = 9$ min $\rightarrow$ speed = $2 \div 9 = 0.222$ km/min = 13.3 km/h. Runner A's sprint is slightly stronger — and crucially, Runner A had already overtaken Runner B. (e) The crossing point: On the graph, the two curves cross somewhere between km 30 and km 35. At km 30, Runner B is at position 30 km but has used 138 min; Runner A is also at 30 km but has used only 132 min — meaning Runner A passed the 30 km mark 6 minutes EARLIER. So although Runner B's displacement line is above Runner A's when you look at position at the same time, by km 30 Runner A is ahead in TIME. The correct interpretation: if you read the graph horizontally (at a fixed time, say $t = 132$ min), Runner A is at km 30 while Runner B has not yet reached km 30. Runner A is ahead. The crossing point on the graph (where both runners have the same displacement at the same time) occurred earlier — around km 22–25 — which is the exact moment Runner A overtook Runner B. After that point, Runner A's curve is always above Runner B's curve, meaning Runner A is always further along the course at any given time. This is exactly the power of a displacement-time graph: it shows you not just where each runner finished, but every moment of relative position throughout the race.
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Section 3 — Velocity-Time Graphs: Drawing, Reading, and Calculating Area

Use the speed values you calculated in Section 2 (one speed value per 5 km segment for each runner) to answer the following. (a) Sketch a velocity-time graph for BOTH runners on the same axes. Plot the average speed for each 5 km segment at the midpoint time of that segment. Label all axes with correct quantities and units. (b) What does the GRADIENT of a velocity-time graph represent? State the formula and units. Is Runner A accelerating or decelerating in any segment? What about Runner B? (c) What does the AREA under a velocity-time graph represent? State the formula and units.	EXEMPLAR RESPONSE: (a) Velocity-time graph description: Horizontal axis: Time (min), from 0 to 200 min. Vertical axis: Speed (km/min) or (km/h), from 0 upward. Approximate segment speeds for each runner (km/min), using $\Delta \text{distance} \div \Delta \text{time}$: Runner A: Each segment $\approx 5 \text{ km} \div 22 \text{ min} \approx 0.227 \text{ km/min}$ — nearly constant across all segments $\rightarrow$ the velocity-time graph is nearly a horizontal straight line. Runner B: Segment 1 (0–5 km): $5 \div 19 = 0.263 \text{ km/min}$; Segment 2 (5–10): $5 \div 21 = 0.238$; Segment 3 (10–15): $5 \div 23 = 0.217$; Segment 4 (15–20): $5 \div 24 = 0.208$; Segment 5 (20–25): $5 \div 26 = 0.192$; Segment 6 (25–30): $5 \div 25 = 0.200$; Segment 7 (30–35): $5 \div 25 = 0.200$; Segment 8 (35–40): $5 \div 23 = 0.217$; Segment 9 (40–42): $2 \div 10 = 0.200 \text{ km/min}$. Runner B's graph: starts high, curves downward to a minimum around km 20–25, then rises slightly for the sprint — a U-shaped or 'swooping' pattern. (b) Gradient of a velocity-time graph = Acceleration Formula: $a = \Delta v \div \Delta t$ (change in velocity divided by change in time) Units: km/min² or m/s² Runner A: The velocity is nearly constant across all segments $\rightarrow$ gradient $\approx 0 \rightarrow$ acceleration ≈ 0. Runner A runs at nearly uniform velocity. This matches the nearly straight displacement-time graph from Section 2.
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(d) Shade and calculate the area under Runner A's velocity-time graph for the first 30 km. Explain what this area represents physically. (e) A matatu on the Nakuru highway accelerates uniformly from rest to 80 km/h in 20 seconds, travels at constant 80 km/h for 5 minutes, then brakes uniformly to a stop in 15 seconds. Sketch the velocity-time graph and calculate: (i) the acceleration during the first phase, (ii) the deceleration during the braking phase, (iii) the total distance travelled.	Runner B: From segment 1 to segment 5, velocity decreases → negative gradient → Runner B is decelerating (slowing down). From segment 7 to segment 8, velocity increases slightly → positive gradient → Runner B is accelerating (the sprint finish). The largest deceleration is in the early-to-middle transition. (c) Area under a velocity-time graph = Distance (displacement) travelled Formula: Area = velocity × time (for uniform velocity rectangles) or $\frac{1}{2} \times \text{base} \times \text{height}$ (for triangular acceleration/deceleration phases) Units: km/min × min = km, or m/s × s = m (d) Area under Runner A's graph for the first 30 km: Runner A's average speed ≈ 0.227 km/min (nearly constant). Time to reach km 30 = 132 min. Area = speed × time = 0.227 km/min × 132 min = 29.96 km ≈ 30 km ✓ Physical meaning: This area represents the total distance Runner A has covered in the first 132 minutes of the race — exactly 30 km. This confirms that the area under a velocity-time graph gives us displacement, linking directly back to the displacement-time graph. The fact that we recover 30 km from the area calculation validates our graph and our understanding of the relationship between the two graph types. (e) Nakuru highway matatu — three-phase journey: Phase 1 — Acceleration from rest to 80 km/h in 20 s: Convert: 80 km/h = $80 \times 1000 \div 3600 = 22.22 \text{ m/s}$; initial velocity $u = 0 \text{ m/s}$. (i) Acceleration = $\Delta v \div \Delta t = (22.22 - 0) \div 20 = 1.11 \text{ m/s}^2$ ✓ Distance in Phase 1 = area of triangle = $\frac{1}{2} \times 20 \text{ s} \times 22.22 \text{ m/s} = 222.2 \text{ m}$ Phase 2 — Constant 80 km/h = 22.22 m/s for 5 minutes = 300 s: Distance = $22.22 \times 300 = 6666 \text{ m} = 6.67 \text{ km}$ Phase 3 — Braking from 22.22 m/s to rest in 15 s: (ii) Deceleration = $\Delta v \div \Delta t = (0 - 22.22) \div 15 = -1.48 \text{ m/s}^2$ (magnitude: 1.48 m/s²) ✓ Distance in Phase 3 = area of triangle = $\frac{1}{2} \times 15 \text{ s} \times 22.22 \text{ m/s} = 166.7 \text{ m}$ (iii) Total distance = $222.2 + 6666 + 166.7 = 7054.9 \text{ m} \approx 7.05 \text{ km}$ ✓ Velocity-time graph: starts at (0, 0), rises in a straight line to (20 s, 22.22 m/s) [Phase 1 — triangle], stays horizontal at 22.22 m/s from t = 20 s to t = 320 s [Phase 2 — rectangle], then drops in a straight line from (320 s, 22.22 m/s) to (335 s, 0) [Phase 3 — triangle]. All axes are labelled: x-axis 'Time (s)', y-axis 'Velocity (m/s)'.
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Section 4 — Answering the Driving Question: The Full Story of Motion

Now bring everything together. Write a full, connected explanation (minimum 3 paragraphs) that directly answers the driving question:

'How can we use mathematics — calculations and graphs — to fully describe and explain the motion of any moving body, from a Nairobi Marathon runner to a matatu on the Nakuru highway?'

Your answer MUST:

- Reference specific calculations you did in Sections 1–3 (use actual numbers)
- Explain the role of BOTH the displacement-time graph AND the velocity-time graph, and what each one reveals that the other does not
- Explain why average speed alone is insufficient and what instantaneous velocity adds to our understanding
- Connect back to the original puzzle: Runner B was ahead at km 30 but Runner A won — use mathematics to explain exactly how this was possible
- Generalise your answer: explain how the same mathematical tools apply to the matatu on the Nakuru highway or to any other moving body

EXEMPLAR RESPONSE:

Mathematics gives us a precise, complete language for describing motion — one that goes far beyond simply saying 'Runner A won' or 'the matatu went fast.' In this unit, we learned that any moving body can be described using two key measurements: displacement (how far it has moved from its starting point, in a specific direction) and velocity (how quickly that displacement changes over time). By calculating average speed — for example, Runner A's average speed of $42 \text{ km} \div 185 \text{ min} = 0.227 \text{ km/min}$ = 13.62 km/h and Runner B's $42 \text{ km} \div 196 \text{ min} = 12.86 \text{ km/h}$ — we can confirm who covered the ground faster overall. But these two numbers alone do not explain the most interesting part of the race: how Runner B was ahead at km 30 and still lost. That deeper explanation requires graphs.

The displacement-time graph tells the story of WHERE each body is at every moment in time. When we plotted both runners on the same axes, the graph revealed something that the split-time table only hinted at: Runner B's curve was higher (further along the course) than Runner A's curve in the early stages, but the gradient of Runner B's curve — which represents velocity — gradually became shallower from km 15 onwards, meaning Runner B was slowing down. Runner A's nearly straight-line curve maintained a consistent gradient of approximately 0.227 km/min throughout. The exact moment the two curves crossed — around km 22–25 — is the precise instant Runner A overtook Runner B. After that crossing point, Runner A's curve is always above Runner B's, meaning Runner A was always further ahead. No single average-speed calculation could show us this crossing point; only the graph reveals it. This is the fundamental power of a displacement-time graph: the shape of the curve carries information that numbers alone cannot.

The velocity-time graph adds a second layer of understanding by showing HOW FAST each body is moving at every moment, and whether it is accelerating or decelerating. For Runner B, the velocity-time graph showed a high starting speed (~0.263 km/min), a declining curve through the middle of the race (reaching a minimum near km 20–25), and a slight upturn for the sprint finish. The gradient of this curve is acceleration — Runner B's negative gradient in the middle confirmed deceleration; the positive gradient near the end confirmed the sprint acceleration. The area under the velocity-time curve confirmed our displacement calculations: for Runner A, the area (speed $\times$ time) over 132 minutes correctly recovered approximately 30 km — physically, the area represents the total distance covered. These same tools apply universally: for the Nakuru highway matatu that accelerated from 0 to 22.22 m/s in 20 seconds (acceleration = 1.11 m/s^2), cruised for 5 minutes, then braked to rest in 15 seconds (deceleration = 1.48 m/s^2), the velocity-time graph gave us the total journey distance of 7.05 km from pure area calculations — no ruler needed on the road. Whether the moving body is an elite Kenyan marathon runner, a matatu, a cyclist in Kisumu, or a ball rolling down a hill in the Rift Valley, the same displacement-time and velocity-time graphs, the same gradient and area calculations, and the same distinction between average and instantaneous velocity give us the complete, precise, and beautiful mathematical description of motion.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Conceptual Understanding: Definitions and Distinctions	All key terms (distance, displacement, speed, velocity, average vs. instantaneous, uniform vs. non-uniform) are defined accurately with correct SI units and clearly distinguished from each other. The student explains, with specific numbers, why a single average speed is insufficient to describe the marathon race.	Most key terms are correctly defined with units. Scalar/vector distinction is present. The student explains why average speed is insufficient, though the explanation may lack specific numerical support.	Some terms are defined but with errors or missing units. Scalar/vector distinction is missing or confused. The student does not clearly explain the limitation of average speed.
Displacement-Time Graph: Construction and Interpretation	Graph is correctly drawn with both runners plotted, axes fully labelled with quantities and units, data points accurate, and curves smooth. The student accurately identifies gradient as velocity, correctly calculates average speeds for both runners, identifies at least three key differences between the curves with segment-by-segment speed calculations, and explains precisely when and why Runner A overtook Runner B using the crossing point.	Graph is drawn with both runners and labelled axes. Gradient is correctly identified as velocity. Average speeds are correctly calculated. At least two differences between curves are identified. The overtaking moment is explained, though the explanation of the crossing point may be imprecise.	Graph is attempted but has unlabelled axes, missing data points, or only one runner plotted. Gradient may be incorrectly identified. Average speed calculations contain errors. The overtaking moment is not clearly explained.
Velocity-Time Graph: Construction, Gradient, and Area	Velocity-time graph for both runners is correctly sketched with labelled axes and segment speeds plotted. Gradient is correctly identified as acceleration with correct formula and units, and acceleration/deceleration of Runner B is correctly described. Area is correctly identified as distance/displacement. The area under Runner A's graph for 30 km is correctly calculated and physically interpreted. The matatu problem (all three phases) is fully and correctly solved with accurate acceleration, deceleration, and total distance calculations, supported by a correct velocity-time graph sketch.	Velocity-time graph is sketched with labelled axes. Gradient is identified as acceleration. Area is identified as distance. Runner A's area calculation is attempted with minor errors. The matatu problem is mostly correct — at least two of the three calculations (acceleration, deceleration, total distance) are correct.	Velocity-time graph is attempted but gradient or area is incorrectly identified. Area calculation is missing or incorrect. The matatu problem is attempted but contains significant errors in at least two of the three phases.
Mathematical Accuracy and Working	All calculations are shown in full with correct formulas stated, substitution shown, and units carried through each step. Final answers are correct and units are stated. Conversions between km/h	Most calculations are shown with formulas and substitution. Minor arithmetic or unit-conversion errors appear but do not consistently undermine the logic. Final answers are mostly	Calculations are incomplete, lack shown working, or contain repeated errors. Formulas are sometimes missing. Units are frequently omitted or incorrect. Conversions between units are mostly

	and m/s, and between minutes and seconds, are handled accurately throughout.	correct with units stated.	absent or wrong.
Synthesis: Answering the Driving Question	The final explanation is a well-structured, multi-paragraph response that directly and fully answers the driving question. It integrates specific calculations from all sections, clearly explains the distinct role of each graph type, explicitly resolves the Runner B paradox using mathematics, and convincingly generalises the tools to the matatu and other moving bodies. The writing is clear, logical, and uses correct mathematical language throughout.	The final explanation addresses the driving question and references calculations from at least two sections. The roles of both graphs are mentioned. The Runner B paradox is addressed, though the mathematical explanation may be partial. Some generalisation to other contexts is present.	The final explanation is vague or general, with little or no reference to specific calculations. Only one graph type is discussed. The Runner B paradox is not resolved mathematically. No generalisation to other moving bodies is attempted.